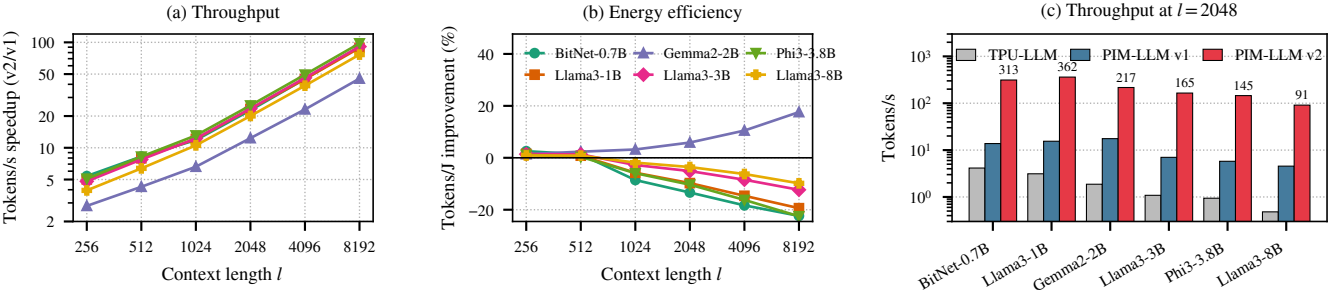
(a) Throughput
Tokens/s speedup (v2/v1)
100
50
20
10
5
2
256
512
1024
2048
4096
8192
Context length l
(b) Energy efficiency
Tokens/J improvement (%)
40
20
0
−20
BitNet-0.7B
Gemma2-2B
Phi3-3.8B
Llama3-1B
Llama3-3B
Llama3-8B
256
512
1024
2048
4096
8192
Context length l
(c) Throughput at $l = 2048$
Tokens/s
10^3
10^2
10^1
10^0
TPU-LLM
PIM-LLM v1
PIM-LLM v2
313
362
217
165
145
91
BitNet-0.7B
Llama3-1B
Gemma2-2B
Llama3-3B
Phi3-3.8B
Llama3-8B